

Improving the ATP Tour

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ACE Advisors: Our Purpose

Elevate the Tour, Elevate the level of tennis worldwide.

Analyze data from the current ATP Tour to understand success metrics and disparities among countries

Discover opportunities for improvement in current player performance and tournament management

Make recommendations to raise performance levels therefore fueling greater engagement in the sport

Background

ATP Tour: Professional men's tennis circuit

Year-long worldwide tour with minimal breaks

Weekly **rankings** based on point totals for that year of competition – constant updated reflection of player performance



Series of Tournaments varying in

- Prize Money
- Points Awarded
- Surface Type (clay, grass, hard court)

Grand Slam Tournaments

- Australian Open
- French Open (Roland Garros)
- Wimbledon
- US Open

Clay Court Season - January to April
August to October

Hard Court Season - April to June
Grass Court Season - June to July

Our Dataset: ATP Matches 2022

General Match Variables:

surface type
winner name
winner country (3 letter code)
winner age
loser age
minutes

Variables Specific to Winner & Loser of each match:

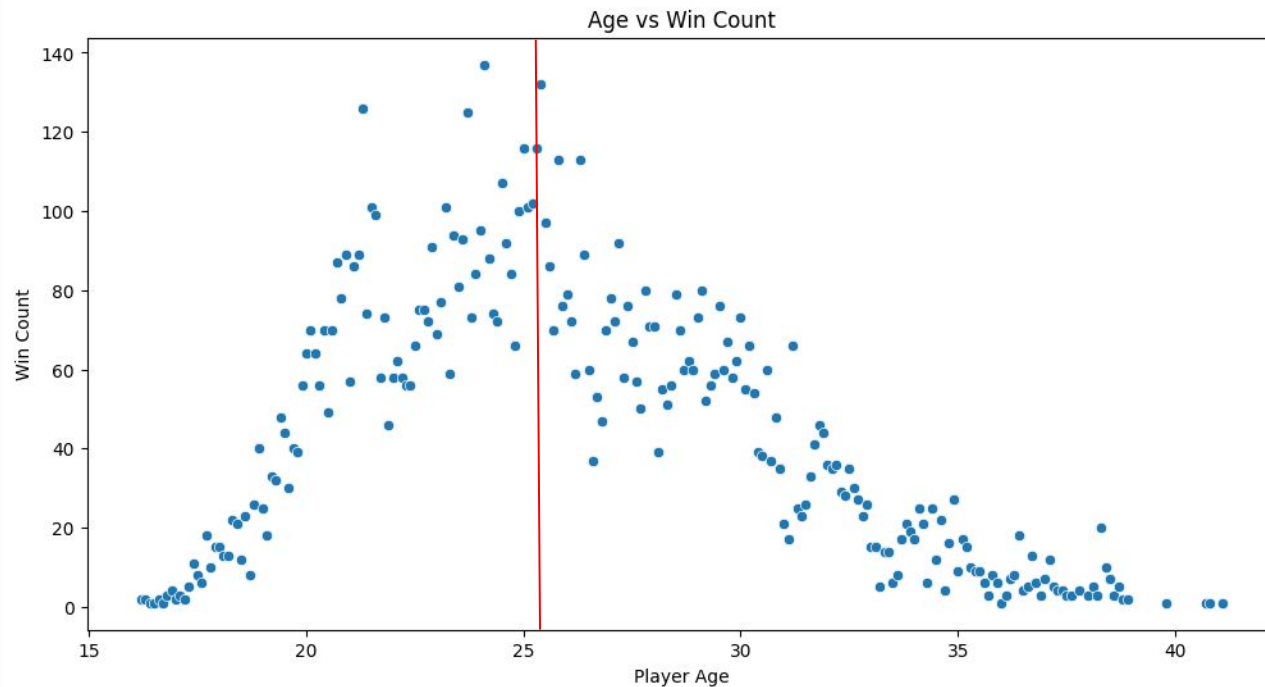
number of aces
number of doubles faults
number of serve points
number of first serves made
number of first-serve points won
number of second-serve points won
number of serve games
number of break points saved
number of break points faced

568,119 observations

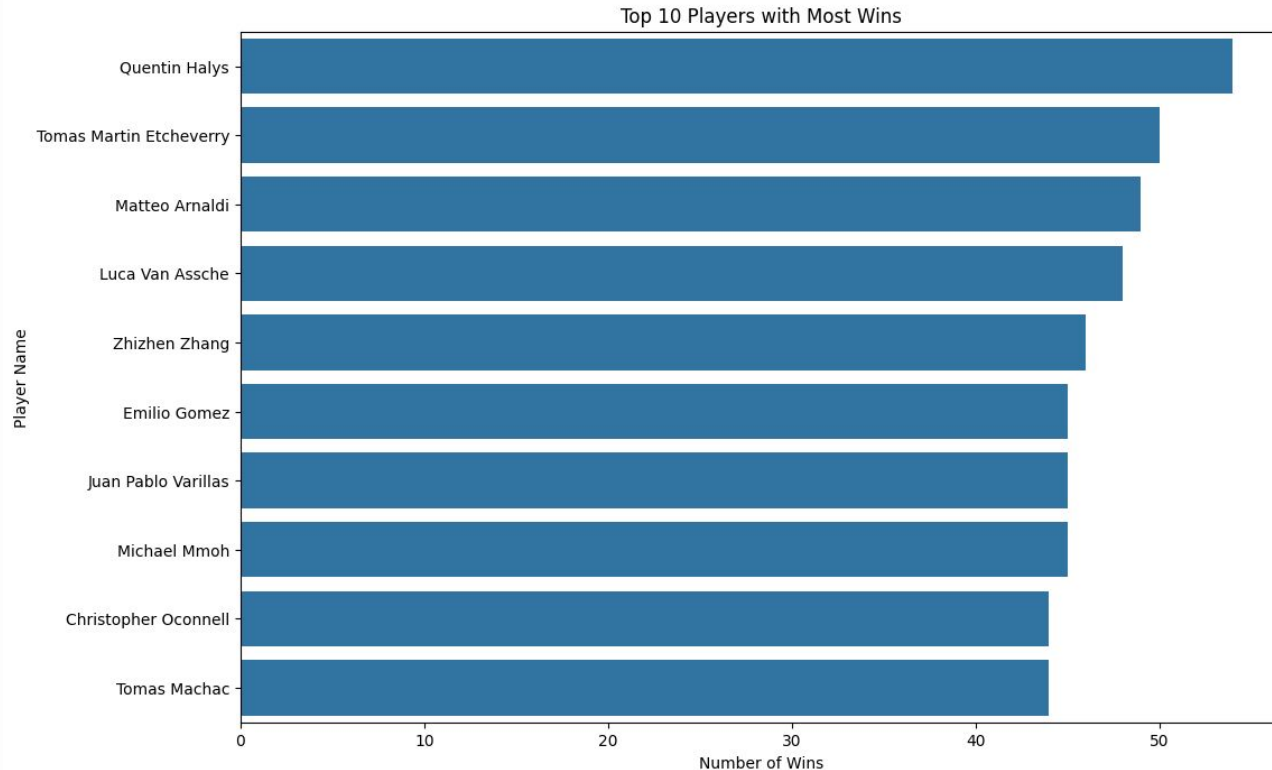
*Spanning all countries, players,
tournaments and matches*

Chose the most recent complete ATP Tour (2022) data to
evaluate accurate opportunities for future growth.

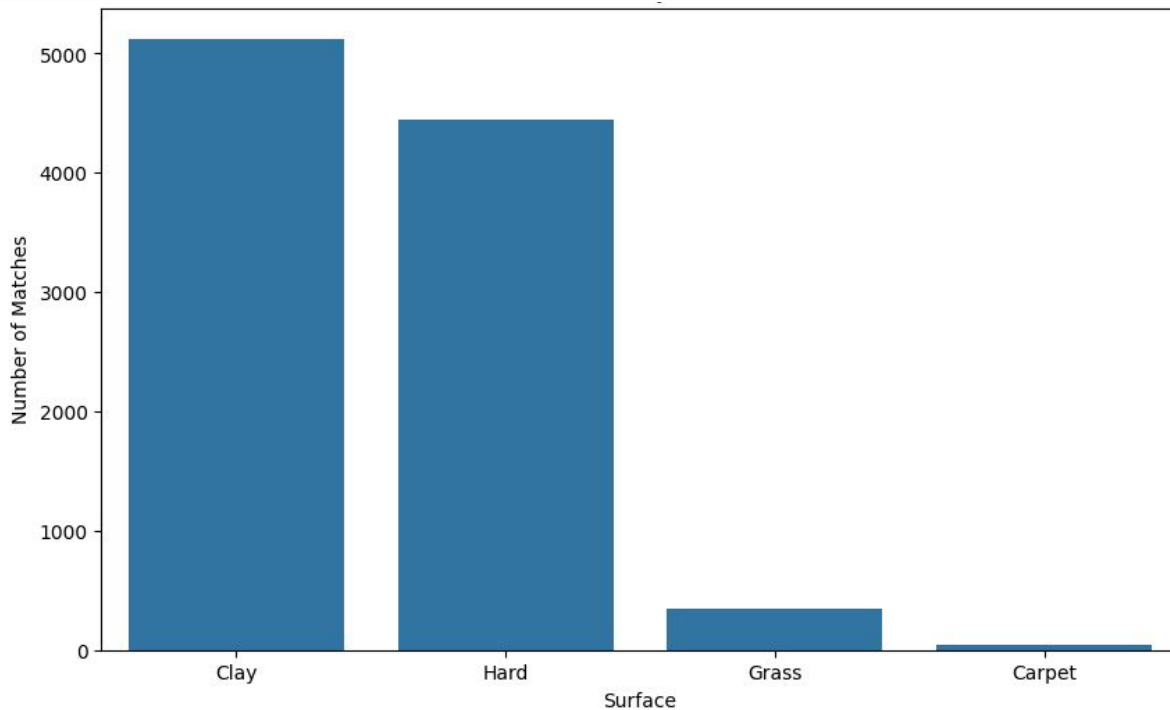
Peak Age in Number of Wins



Most Winning Players in the Tour



Matches Broken Down by Surface Type



Predicting Upsets by Surface Type

Hard Court

$R^2: 0.0534$



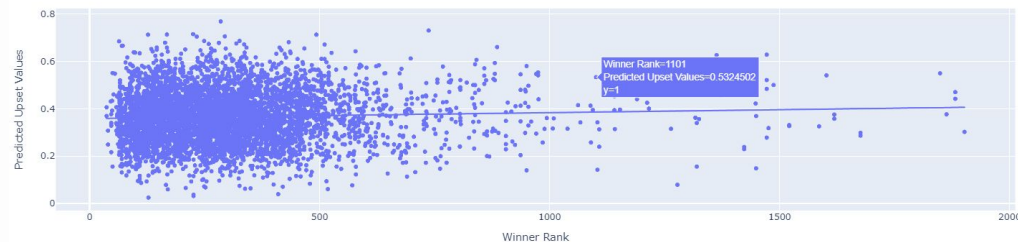
Grass Court

$R^2: 0.1315$



Clay Court

$R^2: 0.0508$



Questions to answer



What are the weak areas of each country?

What variables and factors are limiting a country's success



How can countries maximize their players?

What time are players peaking during their competitive lifespan and how can nations leverage training regimens to optimize their peak years



How can more inexperienced players win?

Significant variables federations can focus on that lead to a players success

**Following our initial analysis, we will conduct
a case study.**

Subject: Spanish Players



Recommendations

Invest more **money** into facilities (Grass and Hard Court)

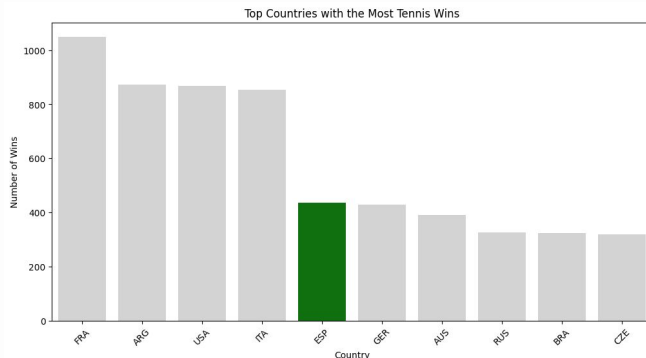
Shift in **training** and **recovery regimen** for players who have surpassed prime age.

Increase various **performance metrics** to expand likelihood of upsetting higher ranked players.

Investing Money in More Facilities

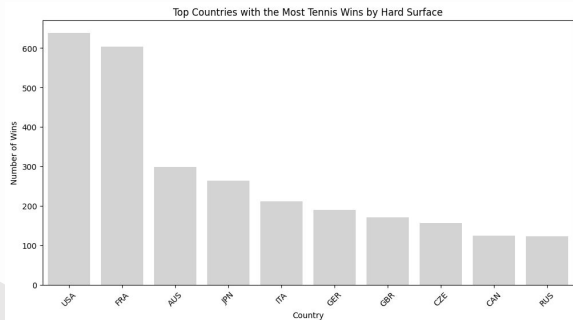
In the Top 5, how can Spain break into the Top 3?

Improving player performance on ALL surface types (especially hard and grass) through training and new infrastructure.

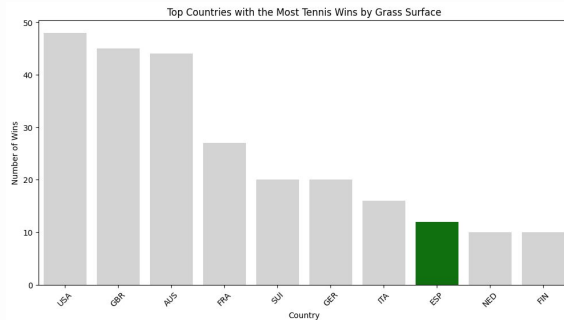


Practice makes perfect!

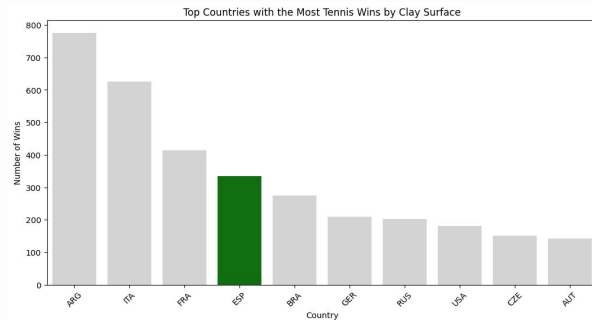
- Investing capital into facilities
- Consistent balanced training on different surface types
- Focusing on *hard and grass* closer to their seasons



Hard Court



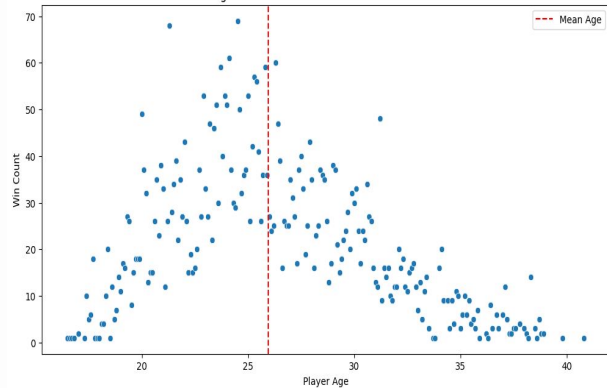
Grass Court



Clay Court

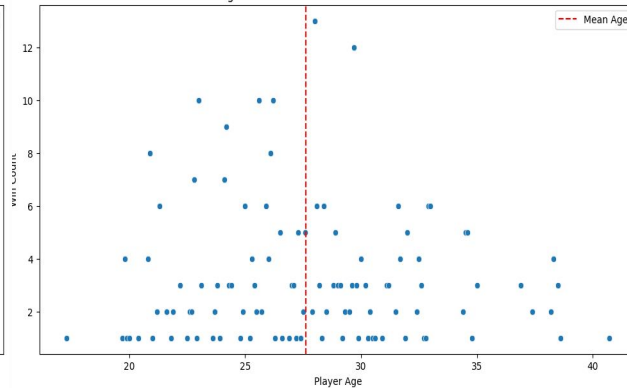
Player Primes by Surface

Age vs Win Count with Mean Reference Line



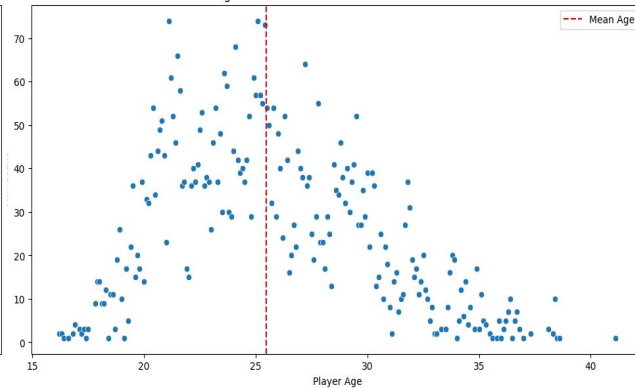
Hard Court
Prime Age: **25.48**

Age vs Win Count with Mean Reference Line



Grass Court
Prime Age: **27.63**

Age vs Win Count with Mean Reference Line



Clay Court
Prime Age: **25.94**

Serve Consistency

	coef	std err	t	P> t	[0.025	0.975]
const	0.4122	0.075	5.528	0.000	0.266	0.558
winner_age	-0.0097	0.002	-6.061	0.000	-0.013	-0.007
loser_age	0.0097	0.002	6.313	0.000	0.007	0.013
minutes	0.0009	0.001	1.723	0.085	-0.000	0.002
w_ace	-0.0032	0.002	-1.522	0.128	-0.007	0.001
w_df	-0.0069	0.004	-1.673	0.094	-0.015	0.001
w_svpt	0.0095	0.003	3.446	0.001	0.004	0.015
w_1stIn	0.0020	0.003	0.733	0.460	-0.003	0.008
w_1stWon	-0.0138	0.005	-2.930	0.003	-0.023	-0.005
w_2ndWon	-0.0138	0.005	-2.873	0.004	-0.023	-0.004
w_SvGms	-0.0327	0.013	-2.553	0.011	-0.058	-0.008
w_bpSaved	-0.0077	0.013	-0.596	0.551	-0.033	0.018
w_bpFaced	0.0052	0.012	0.418	0.676	-0.019	0.030
l_ace	-0.0012	0.002	-0.498	0.618	-0.006	0.004
l_df	0.0040	0.004	1.104	0.270	-0.003	0.011
l_svpt	-0.0043	0.003	-1.556	0.120	-0.010	0.001
l_1stIn	-0.0026	0.002	-1.124	0.261	-0.007	0.002
l_1stWon	0.0075	0.005	1.580	0.114	-0.002	0.017
l_2ndWon	0.0064	0.005	1.302	0.193	-0.003	0.016
l_SvGms	0.0325	0.013	2.495	0.013	0.007	0.058
l_bpSaved	0.0248	0.013	1.906	0.057	-0.001	0.050
l_bpFaced	-0.0248	0.012	-2.006	0.045	-0.049	-0.001

Hard Court

	coef	std err	t	P> t	[0.025	0.975]
const	0.6388	0.262	2.441	0.015	0.124	1.154
winner_age	-0.0154	0.006	-2.564	0.011	-0.027	-0.004
loser_age	0.0074	0.006	1.307	0.192	-0.004	0.019
minutes	-0.0016	0.003	-0.505	0.614	-0.008	0.005
w_ace	-0.0083	0.007	-1.127	0.261	-0.023	0.006
w_df	-0.0062	0.016	-0.386	0.700	-0.038	0.025
w_svpt	0.0152	0.010	1.463	0.144	-0.005	0.036
w_1stIn	-0.0121	0.011	-1.121	0.263	-0.033	0.009
w_1stWon	-0.0112	0.020	-0.571	0.569	-0.050	0.028
w_2ndWon	-0.0222	0.020	-1.106	0.270	-0.062	0.017
w_SvGms	-0.0121	0.047	-0.258	0.797	-0.105	0.081
w_bpSaved	-0.0371	0.054	-0.685	0.494	-0.143	0.069
w_bpFaced	0.0476	0.053	0.897	0.371	-0.057	0.152
l_ace	-0.0059	0.009	-0.690	0.491	-0.023	0.011
l_df	0.0147	0.014	1.013	0.312	-0.014	0.043
l_svpt	-0.0178	0.009	-1.879	0.061	-0.036	0.001
l_1stIn	0.0152	0.009	1.738	0.083	-0.002	0.032
l_1stWon	-0.0007	0.017	-0.039	0.969	-0.035	0.034
l_2ndWon	0.0173	0.018	0.941	0.347	-0.019	0.054
l_SvGms	0.0857	0.049	1.753	0.080	-0.010	0.182
l_bpSaved	0.0420	0.051	0.817	0.415	-0.059	0.143
l_bpFaced	-0.0456	0.048	-0.957	0.339	-0.139	0.048

Grass Court

	coef	std err	t	P> t	[0.025	0.975]
const	0.3679	0.071	5.148	0.000	0.228	0.508
winner_age	-0.0117	0.002	-7.356	0.000	-0.015	-0.009
loser_age	0.0127	0.001	8.549	0.000	0.010	0.016
minutes	0.0001	0.000	0.273	0.785	-0.001	0.001
w_ace	0.0032	0.003	1.225	0.220	-0.002	0.008
w_df	-0.0005	0.004	-0.123	0.907	-0.008	0.007
w_svpt	0.0114	0.003	4.148	0.000	0.006	0.017
w_1stIn	-0.0037	0.002	-1.536	0.125	-0.008	0.001
w_1stWon	-0.0142	0.005	-2.823	0.005	-0.024	-0.004
w_2ndWon	-0.0185	0.005	-3.540	0.000	-0.029	-0.008
w_SvGms	0.0024	0.014	0.172	0.863	-0.025	0.029
w_bpSaved	-0.0041	0.014	-0.293	0.769	-0.031	0.023
w_bpFaced	0.0020	0.013	0.149	0.881	-0.024	0.028
l_ace	0.0063	0.003	2.007	0.045	0.000	0.012
l_df	-0.0071	0.003	-2.190	0.029	-0.013	-0.001
l_svpt	-0.0054	0.003	-1.997	0.046	-0.011	-0.000
l_1stIn	-0.0024	0.002	-1.222	0.222	-0.006	0.001
l_1stWon	0.0179	0.005	3.559	0.000	0.008	0.028
l_2ndWon	0.0135	0.005	2.591	0.010	0.003	0.024
l_SvGms	-0.0182	0.014	-1.310	0.190	-0.045	0.009
l_bpSaved	-0.0147	0.014	-1.030	0.303	-0.043	0.013
l_bpFaced	0.0125	0.013	0.924	0.355	-0.014	0.039

Clay Court

Overview & Next Steps



Investing in different training facilities based on weakest court type: Hard and Grass

Shift in training regimen, once player is past peak age of 25

Optimize serve consistency across all surface types to fuel upsets

Thank You!

Any questions?